

Rahul Kallidan

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Professional Summary

Senior Mechanical Engineer (Robotics) with 7+ years designing and launching Industrial AMRs and commercial cleaning robots. Experienced in developing sensor-enabled mechanisms—including tactile safety bumpers, force-driven lifting modules, and IP-sealed sensor layouts—for robust field deployment. Delivered 25% unit cost reduction and 33% faster assembly through DFM/DFA, tolerance stack-ups, and supplier optimisation. Shipped 1.8k+ robots to global markets with validated ingress protection (IEC 60529 IPX3), FEA/CFD analyses, and manufacturable CAD to production drawing workflows. Passionate about building scalable, reliable, and intelligent robotic hardware for embodied AI applications.

Core Skills

- **Mechanical Design:** DFM/DFA, GD&T (ASME Y14.5), tolerance stacks, value engineering, design reviews
- **Manufacturing:** CNC, injection molding, die casting, vacuum casting, sheet metal, jigs/fixtures, Value Analysis & Value Engineering (VAVE)
- **Robotics/Systems:** AMR architecture, suspensions, lifting/tug modules, cleaning mechanisms, docking stations, IP sealing, sensor configuration
- **Analysis/Software:** SolidWorks PDM/3DEXperience, Ansys (FEA/CFD), Matlab, ROS, SolidWorks Composer
- **Quality & Compliance:** DFMEA, ECNs, IQC/OQC, IEC 60529 (IPX3), ISO 12100, ISO 45001
- **Project & Ops:** Agile workflows (Confluence, Notion), BOM & cost engineering, cross-team collaboration, ECN, supplier coordination

Experience

Senior Mechanical Engineer, Lionsbot International Pte Ltd – Singapore **Apr 2022 – Jul 2025**

- Owned mechanical design and system architecture for **R3 Scrub Pro** from concept to production; shipped 1.8k+ units across multiple markets, achieving IPX3 ingress compliance and sensor enabled safety.
- Delivered 25% part cost reduction via design consolidation and supplier process changes; reduced assembly time 33% through DFA and fixture redesign, enabling manufacturable scaling.
- Designed and validated handlebar, cleaning, vacuum chamber, and tactile bumper subsystems, integrating safety sensing and force-driven cleaning mechanisms.
- Managed cross-functional timelines, supplier handovers, and production SOPs, enabling smooth transition from prototype builds to mass manufacturing; coordinated design → validation → ramp with QA and production teams.
- Developed suspension (12° slope capability) and chassis design using first-principles engineering (traction, spring, and stability calculations); completed CAD integration, sensor layouts, and fluid systems.

Senior Mechanical Engineer, Botsync Technologies – Bangalore, India **Jul 2021 – Mar 2022**
Mechanical Engineer, Botsync Technologies – Bangalore, India **Mar 2020 – Jun 2021**

- Sole owner of all mechanical development for **MAG500 and MAG1000** AMRs (500 kg & 1000 kg payloads), from architecture and subsystem design to validation and production release.
- Designed chassis, suspension, pallet lifters, tug modules, and docking stations, integrating force-sensing modules and alignment features for reliable logistics automation.
- Led CAD modelling, tolerance stack-ups, and FEA validation, ensuring structural robustness and repeatable manufacturability.
- Established PDM workflows, BOM traceability, and ECN processes, reducing design cycle time and ensuring quality compliance.
- Owned supplier engagement and prototype builds, translating designs into manufacturable assemblies.
- Partnered with suppliers for prototype builds and pilot deployments; validated sensor-enabled lifting and docking systems at client sites (e.g., TVS Motor, Ford Manufacturing), resolving CAPAs within 2 weeks.

Solution Design Engineer, Addverb Technologies – Noida, India**Jan 2019 – Feb 2020**

- Owned mechanical design for AMR pilot projects with FMCG clients (e.g., HUL Ice Cream, Lifebuoy, Cofresh); delivered functional prototypes and BOM packages that achieved 100% first-pass site acceptance.
- Designed conveyor systems (belt and roller) with integrated safety and control sensors, optimising layouts for throughput and reliability.
- Applied GD&T and tolerance analysis in manufacturing drawings, ensuring supplier clarity and manufacturability.
- Commissioned conveyor systems at client sites, achieving continuous operation and client sign-off.
- Collaborated with cross-functional controls/software teams to validate sensor-enabled automation systems in production environments.

Mechanical Engineer, Airlitix Inc – California (Remote Work)**Nov 2017 – Dec 2018**

- Performed CFD simulations in Ansys to evaluate duct geometries for greenhouse drone applications, selecting designs that maximised thrust efficiency while minimising losses.
- Contributed to prototype drone design, including CAD modelling, ducted fan integration, and proof-of-concept builds.
- Collaborated with the computer vision team to define mechanical requirements for sensor placement and early perception validation.
- Delivered design recommendations that improved subsequent drone prototypes for agricultural automation.

Education

Amrita School of Engineering, Master of Technology - Robotics and Automation

Jul 2015 - Jun 2017

Visvesvaraya Technological University, Bachelor of Engineering - Mechanical Engineering

Jul 2011 - Jun 2015

Publications

- Design and validation of low-cost wearable assistive device for carrying back loads, IConAMMA-2017
- Low cost mechanical assistive aid for workers carrying head loads, TIAR 2017 (IEEE Xplore)